

Polynomial Regression

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

```
In [2]: dataset = pd.read_csv(r"C:\Users\sande\OneDrive\Desktop\Naresh IT\data science and dataset")
```

```
Out[2]:
```

	Position	Level	Salary
0	Jr Software Engineer	1	45000
1	Sr Software Engineer	2	50000
2	Team Lead	3	60000
3	Manager	4	80000
4	Sr manager	5	110000
5	Region Manager	6	150000
6	AVP	7	200000
7	VP	8	300000
8	CTO	9	500000
9	CEO	10	1000000

```
In [3]: X = dataset.iloc[:, 1:2].values
y = dataset.iloc[:, 2].values
```

linear model -- linear algor (degree - 1)

```
In [4]: from sklearn.linear_model import LinearRegression
lin_reg = LinearRegression()
lin_reg.fit(X, y)
```

Out[4]:

LinearRegression		
Parameters		
fit_intercept	True	
copy_X	True	
tol	1e-06	
n_jobs	None	
positive	False	

polynomial model (bydefeaut degree - 2)

```
In [6]: from sklearn.preprocessing import PolynomialFeatures
poly_reg = PolynomialFeatures(degree=5)
X_poly = poly_reg.fit_transform(X)

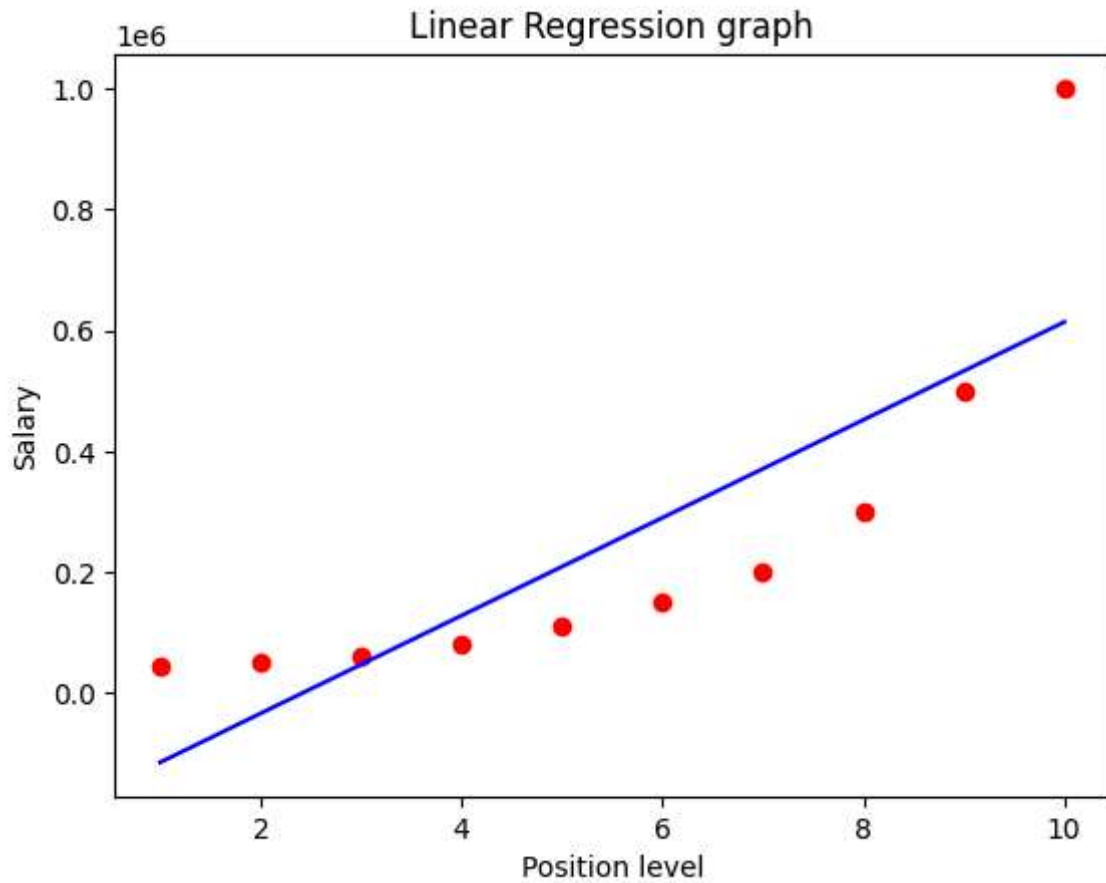
poly_reg.fit(X_poly, y)
lin_reg_2 = LinearRegression()
lin_reg_2.fit(X_poly, y)
```

Out[6]:

LinearRegression		
Parameters		
fit_intercept	True	
copy_X	True	
tol	1e-06	
n_jobs	None	
positive	False	

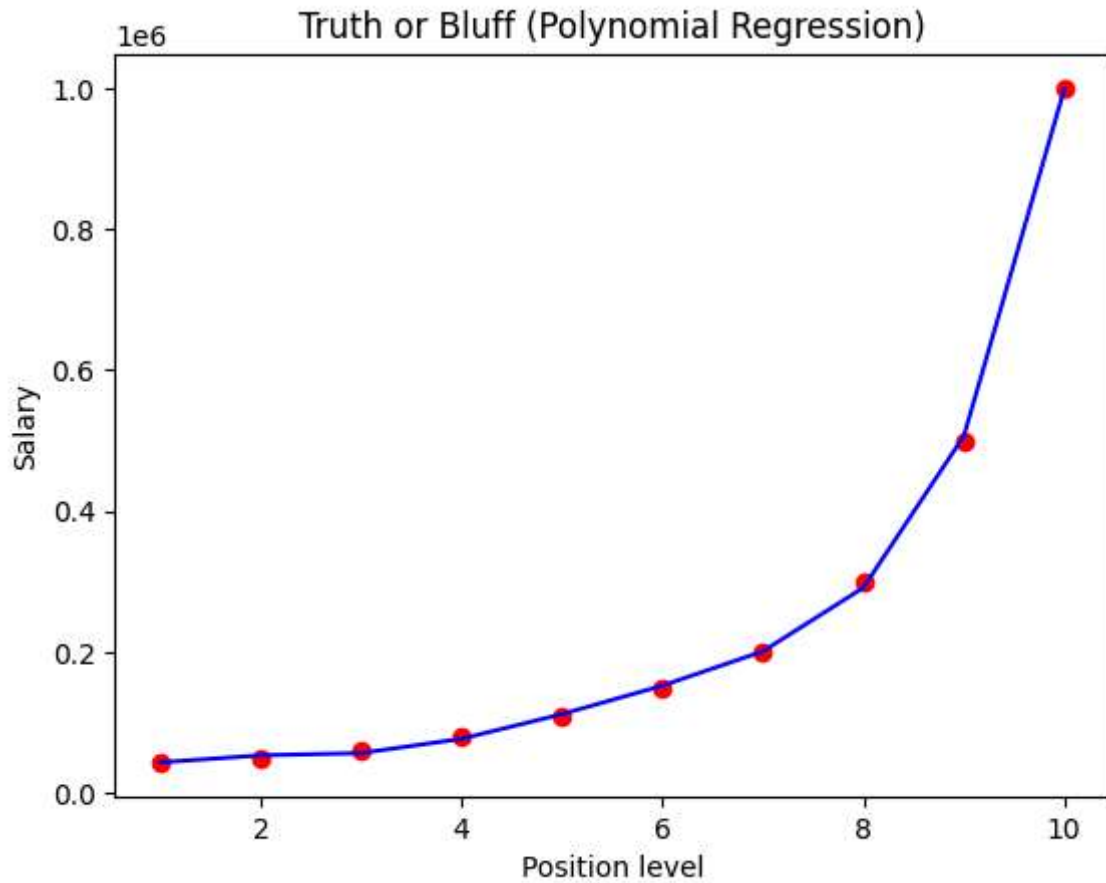
linear regression visualizaton

```
In [7]: plt.scatter(X, y, color = 'red')
plt.plot(X, lin_reg.predict(X), color = 'blue')
plt.title('Linear Regression graph')
plt.xlabel('Position level')
plt.ylabel('Salary')
plt.show()
```



poly nomial visualization

```
In [8]: plt.scatter(X, y, color = 'red')
plt.plot(X, lin_reg_2.predict(poly_reg.fit_transform(X)), color = 'blue')
plt.title('Truth or Bluff (Polynomial Regression)')
plt.xlabel('Position level')
plt.ylabel('Salary')
plt.show()
```



prediction

```
In [9]: lin_model_pred = lin_reg.predict([[6.5]])  
lin_model_pred  
  
poly_model_pred = lin_reg_2.predict(poly_reg.fit_transform([[6.5]]))  
poly_model_pred
```

```
Out[9]: array([174878.07765173])
```